



U.S. Department  
of Transportation

**Pipeline and Hazardous  
Materials Safety Administration**

April 30, 2008

East Building, PHH-30  
1200 New Jersey Avenue S.E.  
Washington, D.C. 20590

DOT-SP 14684  
(SECOND REVISION)

**EXPIRATION DATE: September 30, 2008**

1. GRANTEE: A.O. Smith Corporation  
Milwaukee, WI
2. PURPOSE AND LIMITATION:
  - a. This emergency special permit authorizes the one-way transportation in commerce of approximately 49,300 non-DOT specification cylinders containing compressed air at a pressure not greater than 40 psig by motor vehicle. This special permit provides no relief from the Hazardous Materials Regulations (HMR) other than as specifically stated herein.
  - b. The safety analyses performed in development of this special permit only considered the hazards and risks associated with transportation in commerce. The safety analyses did not consider the hazards and risks associated with consumer use, use as a component of a transport vehicle or other device, or other uses not associated with transportation in commerce.
  - c. No party status will be granted to this special permit.
3. REGULATORY SYSTEM AFFECTED: 49 CFR Parts 106, 107 and 171-180.
4. REGULATIONS FROM WHICH EXEMPTED: 49 CFR § 173.302a in that non-DOT specification cylinders are not authorized, except as provided herein; § 172.301(c) in that the marking requirements are waived, and subpart F of Part 172 in that placards are not required.

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5. BASIS: This emergency special permit is based on the application of A.O. Smith Corporation dated April 24, 2008, submitted in accordance with § 107.117 and a determination that it is necessary to prevent significant economic loss.
6. HAZARDOUS MATERIALS (49 CFR § 172.101):

| Hazardous Materials Description |                           |                               |                  |
|---------------------------------|---------------------------|-------------------------------|------------------|
| Proper shipping name            | Hazard Class/<br>Division | Identi-<br>fication<br>Number | Packing<br>Group |
| Air, compressed                 | 2.2                       | UN1002                        | N/A              |

7. SAFETY CONTROL MEASURES:

a. PACKAGING - Packagings prescribed are approximately 49,300 non-DOT specification water pump system tanks that are manufactured from cold rolled draw quality steel that meets the ASTM 619 or ASTM 620 standard. The tanks have a water capacity of 14 to 116 US gallons and a maximum service pressure of 100 or 125 psig. The maximum tank diameter is 26 inches as shown in the A.O. Smith drawings on file with the Office of Hazardous Materials Special Permits and Approvals (OHMSPA) and the tanks conform with § 173.306(g) except as follows:

- (a) § 173.306(g)(1) - Applies except that the outside diameter may not exceed 26 inches.
- (b) § 173.306(g)(3) - The stress at pressure must not exceed 22,000 psi using the formula:

$$S = Pd/2t$$

Where:

S = wall stress in pounds per square inch;  
P = prescribed pressure for the tank of at least 3 times charged pressure at 70°F or 100 psig, whichever is greater;  
d = inside diameter in inches;  
t = minimum wall thickness, in inches.

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- (c) § 173.306(g)(4) - The burst pressure must be at least 8 times the charge pressure at 70°F.
  - b. TESTING - Each cylinder design must be hydrostatically pressure tested to 300 psig (2070 kPa) or 3 three times the maximum service pressure, whichever is greater, for 15 minutes with no external leakage observed.
  - c. OPERATIONAL CONTROLS -
    - (1) The maximum pressure in each cylinder may not exceed 38 psig during transportation.
    - (2) Transportation is authorized from A.O. Smith Corporation and A.O. Smith Distributors in truckload or less than truckload quantities with no intermediate stops.
- 8. SPECIAL PROVISIONS:
  - a. A person who is not a holder of this special permit who receives a package covered by this special permit may reoffer it for transportation provided no modifications or changes are made to the package or its contents and it is reoffered for transportation in conformance with this special permit and the HMR.
  - b. A current copy of this special permit must be maintained at each facility where the package is offered or reoffered for transportation.
  - c. The marking requirements of § 172.301(c) are waived.
  - d. Shipments are not subject to the placarding requirements of subpart F of Part 172 provided the identification number (UN1002) is marked on the transport vehicle on an orange panel or white square on point configuration on each end and each side (see §§ 172.332 and 172.336, respectively).
- 9. MODES OF TRANSPORTATION AUTHORIZED: Motor Vehicle.
- 10. MODAL REQUIREMENTS: A current copy of this special permit must be carried aboard each motor vehicle used to transport packages covered by this special permit.

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11. COMPLIANCE: Failure by a person to comply with any of the following may result in suspension or revocation of this special permit and penalties prescribed by the Federal hazardous materials transportation law, 49 U.S.C. 5101 et seq.:

- o All terms and conditions prescribed in this special permit and the Hazardous Materials Regulations, 49 CFR Parts 171-180.
- o Persons operating under the terms of this special permit must comply with the security plan requirement in Subpart I of Part 172 of the HMR, when applicable.
- o Registration required by § 107.601 et seq., when applicable.

Each "Hazmat employee", as defined in § 171.8, who performs a function subject to this special permit must receive training on the requirements and conditions of this special permit in addition to the training required by §§ 172.700 through 172.704.

No person may use or apply this special permit, including display of its number, when this special permit has expired or is otherwise no longer in effect.

Under Title VII of the Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (SAFETEA-LU)- 'The Hazardous Materials Safety and Security Reauthorization Act of 2005' (Pub. L. 109-59), 119 Stat. 1144 (August 10, 2005), amended the Federal hazardous materials transportation law by changing the term "exemption" to "special permit" and authorizes a special permit to be granted up to two years for new special permits and up to four years for renewals.

12. REPORTING REQUIREMENTS: Shipments or operations conducted under this special permit are subject to the Hazardous Materials Incident Reporting requirements specified in 49 CFR §§ 171.15 - Immediate notice of certain hazardous materials incidents, and 171.16 - Detailed hazardous

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materials incident reports. In addition, the grantee(s) of this special permit must notify the Associate Administrator for Hazardous Materials Safety, in writing, of any incident involving a package, shipment or operation conducted under terms of this special permit.

Issued in Washington, D.C.:



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for Theodore L. Willke  
Associate Administrator for Hazardous Materials Safety

Address all inquiries to: Associate Administrator for Hazardous Materials Safety, Pipeline and Hazardous Material Safety Administration, U.S. Department of Transportation, East Building PHH-30, 1200 New Jersey Avenue, Southeast, Washington, D.C. 20590.

Copies of this special permit may be obtained by accessing the Hazardous Materials Safety Homepage at [http://hazmat.dot.gov/sp\\_app/special\\_permits/spec\\_perm\\_index.htm](http://hazmat.dot.gov/sp_app/special_permits/spec_perm_index.htm) Photo reproductions and legible reductions of this special permit are permitted. Any alteration of this special permit is prohibited.

PO: BMoore:kah